

MATRUBHOOMI MAHAVIDYALAYA,UDGIR



This is certify that

Miss.Wadgaonkar Vaibhavi Rajkumar

Miss.Deshmukh Shradha Yadavrao

The students of **BCS T.Y. SEM.VI** of this
Mahavidyalaya has successfully completed

The Project On

Timetable Management System

As per the requirements of

Swami Ramanand Thirth Marathwada University, Nanded.

In partial fulfilment of degree of

“Bachelor of Computer Science”

During the Academic Year **2023-2024.**

Examiner

Principal

Prof. Usha Kulkarni

H.O.D

Prof. Ustad S.M.



MATRUBHOOMI MAHAVIDYALAYA ,UDGIR

Department of Computer Science

CERTIFICATE

This is certify that,

Miss.Wadgaonkar Vaibhavi Rajkumar

Miss.Deshmukh Shradha Yadavrao

The students of **BCS T.Y. SEM.VI** Of this Institute has Successfully completed the Project in “**TIMETABLE MANAGEMENT SYSTEM**”
Computer Language naming, As per the requirement of

Swami Ramanand Teerth Marathwada University,Nanded.

In partial fulfillment of degree of **BCS TY** during the Academic Year **2023-2024.**

Seat No:-

Seat No:-

Examiner Name

Guide

Prof.Kulkarni.U.B.

H.O.D.

Prof.Ustad S.M.

A

PROJECT ON

TIMETABLE MANAGEMENT

SYSTEM

ACKNOWLEDGEMENT

The success and final outcome of this project required a lot of guidance and assistance from many people and we are extremely privileged to have got this all along the completion of my project. All that we have done is only due to such supervision and assistance and we would not forget to thank them.

We respect and thank to **Principal Miss. Kulkarni Mam** for providing us on opportunity to do the project work in College Lab and giving us all support and guidance which made us complete the project duty. We extremely thankful to her for providing such a nice support and guidance, although he had busy schedule managing the corporate affairs.

We owe our deep gratitude to our **Project Guide Miss.Kulkarni Mam** who took keen interest on our project work and guided us all along, till the completion of our project work by providing all the necessary information for developing a good system.

Sir for their encouragement and more over for their timely support and guidance till the completion of our project work. We heartily thank our **H.O.D. Mr Ustad Sir** , for this guidance and suggestions during this project work.

We are thankful to and fortunate enough to get constant encouragement, support and guidance from **Miss.Kulkarni Mam** which helped us in successfully completing our project work. Also, We would like to extend our sincere esteems to all staff in College for their timely support.

Your Sincerely,

Miss.Vaibhavi R.Wadgaonkar

Miss.Shradha Y. Deshmukh

ABSTRACT

The Project Student Result processing Software is a complete Multi-user Software - The Scorer accept and then processes the scores of Students and subsequently produces their report cards and other analysis Charts.

This project aimed at providing a school to analyze The student Performance as well as teacher performance by generating reports like Merit list, Progress reports and Teacher performance reports etc. as key Features to reduce manual analysis.

Thanking you!

Your Sincerely,

Miss.Vaibhavi R.Wadgaonkar

Miss.Shradha Y.Deshmukh

Place:-Udgir

Date:-

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INTRODUCTION

Timetabling concerns all activities with regard to producing a schedule that must be subjective to different constraints. Timetable can be defined as the optimization of given activities, actions or events

to a set of objects in space-time matrix to satisfy of desirable constraints. A key factor in running an educational centre or basically an academic environment is the need for a well-planned, well-throughout and clash-free timetable. Back in the days when technology was not in wide use, (lecture) timetables were manually created by the academic institution.

Every college year, tertiary institutions are faced with the tedious task of drawing up academic timetables that satisfies the various courses and the respective examination being offered by the different departments. Timetable development process starts when each Head of Department provide the following information to be used for timetable scheduling. The information provides the modules with dates, time and venues suitable in a particular semester:

Examinable courses in a particular semester.

Dates for lectures to be held (Lectures can be scheduled between Monday and Friday).

Specified time for lectures (i.e. Between 8am and 4pm)

The venue of the scheduled lectures. A timetabling problem consists of four parameters and they are:

T (set of time),

R (set of available resources),

M (set of scheduled contacts) and

C (set of constraints). This problem assigns time and resources to the contacts on such a way that the constraints will be

satisfied. In various timetabling problems, educational timetabling has been generally examined from

practical standpoint. The quality of the timetable determines the quality of time dedicated by lecturers, students

and administrators to academic activities. **Various academic timetabling includes:**

College timetable

Lecture timetable

Examination timetable and

Assignment timetable.

PROJECT OVERVIEW

Timetabling concerns all activities with regard to producing a schedule that must be subjective to different constraints. Timetable can be defined as the optimization of given activities, actions or events to a set of objects in space-time matrix to satisfy a set of desirable constraints. A college timetable is a temporal arrangement of a set of lectures and classrooms in which all given constraints are satisfied.

Creating such timetables manually is complex and time-consuming process. By automating this process with computer assisted timetable generator can save a lot of time. Mostly, in Sri Lanka, the system of managing a university timetable is done manually. This is a cumbersome task, especially with more constraints in concern. Firstly, the generation of the timetable is largely time-consuming. Secondly, the management (updates) and management reporting is overwhelming.

Thirdly, it is prone to many unexpected “timetable clashes” and inconsistencies. “Web-Based Timetable Management System developed for the technical services department of University of Vocational Technology. It is a semi-government technical training institute that

comes directly under the Ministry of Youth affairs. It consists of five main departments providing various

training programs from certificate to undergraduate level.

The institution provides training for more than 1000 students per year” (UNIVOTEC, 2017). Proposed web-based system will be accessed by all level of staff in Technical Services Unit (TSU) and mainly course coordinators lecturers demonstrators and staff directly involve for the resource allocation process of the university. Univotec Timetable Management System (UVTTMS) allows creating timetable considering following constrains.

1. Course MIT offers maximum 20 credits in semester three within 18 weeks of studies, including one-week mid-semester week and two weeks study leave for end semester examinations.
2. Professor ABC can teach maximum .
- 3.credits per delivery Lecture room R015 is not available on Tuesdays during September-December delivery, every year.
4. A laboratory practical session conducted in L003 shall not exceed one hour. However, it will be available 24X7 from October 2020, if the senate approves the proposal.
5. Number of students enroll for particular module that is deliver in lecturer room or lab shall not exceed its capacity.

This system has been developed in Microsoft® Windows 10 environment using Apache, PYTHON and MySQL three open source technologies that are often combined to develop web applications.

Combination of these technologies ensures a development of error free web-based system which will full fill customer requirement on giventime-line.

PROJECT ANALYSIS

Scope of the Project

The following are the functionalities of the system

- a) Provides an effective and efficient constraint management (add, update, delete) Lecturer, resource, module, time slot, class (batch) management (add, update, delete)
- b) Provides Management Information System reporting on demand. (by date, time slot, lecturer/ resource person/ resource, module, and etc.)
- c) Produces a fully functional timetable generator engine/ algorithm that can automatically cater to each constraint.

Advantages

Colleges are known for being the place that makes us disciplined and teaches us the importance of punctuality. In order to run college management efficiently, it

is essential to have an order in the schedule. Abiding time in a college is necessary to hone the learning experience of the students and to achieve the optimal use of resources.

However, that doesn't absolve the hundreds of hours the staff has to spend on managing timetables for classes and courses, especially when technology is growing every day at a considerable pace and has all the time-saving solutions such as timetable management systems.

- Paperless Timetable Generation
- Clash free schedule generation
- Substitution management
- Simplifying The Payroll Calculation
- Better Security of Data
- Accessible employee details

Top 5 Leading Time Table Management Software for Colleges

Prime Time Table

This college scheduling software is flexible and fun to use. The college timetable class wise and teacher wise software lets

copy paste and importing of data. This class timetable software uses state of the art technology. It also has forbidden slots for teachers.

TimeTabler

This college timetable software is reliable and helps in tracking and scheduling college and college timetables. It is designed with users of different types in mind. The elective subjects of students are considered. This college timetable making software is highly helpful in monitoring the absence of teachers.

EduTimer

The college timetable software helps in formulating the best schedule for teachers. It also creates college timetable class wise and teacher wise software. The college timetable maker software helps with guidelines for new users at any point in time.

AscTimetables

This college timetable software is designed to enter inputs faster and easier. This college timetable generator software is also easy to customize. It has many advanced features. The college timetable system also lets the teachers and the students use the software on their mobile devices too.

Unit Express

This best college timetable software is known for its consistent support offered. Create the timetable you need, and display in any way required. Covering the shortfall of staff is also easy. Overview of the timetable and make changes using this college timetable generator. The institution provides training for more than 1000 students per year” (UNIVOTEC, 2017). Proposed web-based system will be accessed by all level of staff in Technical Services Unit (TSU) and mainly course coordinators lecturers demonstrators and staff directly involve for the resource allocation process of the university. Univotec Timetable Management System (UVTTMS) allows creating timetable considering following constrains.

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Types of timetable management

Different types of timetables that you can perfect using the timetable management system in educational ERP are:

Consolidated Timetable

It aims to help the head of department (HOD) to keep track of a teacher's schedule and ensure any required changes are made immediately.

Class-wise timetable

It allows teachers to plan their day according to the different classes they teach in.

Teacher-wise timetable

It allows teachers to check their classes on a daily, weekly, and monthly basis.

Timetable for sports session

It highlights the schedules for all sports activities to keep teachers and students ready with specific uniforms and equipment.

Co-curricular activities timetable

It allows students to be ready for co-curricular activities at a scheduled time.

Characteristics of Time table management system

The following are some of the perks of using a timetable management system:

Unplanned systems can be taxing: A well-timed routine can help by keeping students and teachers involved and disciplined. It will also save time by lowering the amount of time spent in chaos and confusion.

Faculty distribution is equitable: The timetable management system evenly distributes lectures to each faculty member and prevents work from being distributed based on events.

Rearrangement of the timetable based on circumstances: When necessary, new classes could be added or removed from the timetable.

Event and holiday calendar planner: Holidays and events could be added in the backend which can

further be viewed by the user in the form of a fully furnished yearly calendar.

Workflow of timetable management

Here's how you can plan a smooth workflow for timetable management.

Assign one practical period to multiple teachers so more students can get personal attention in the presence of more teachers.

Schedule classes with variations in the time slots and weekdays.

Schedule multiple periods in a day with fixed or variable duration, depending on the significance of subjects taught.

Make timetables according to the class and the subjects students are supposed to study during the day.

Don't forget to include assembly, lunch breaks, and a small recess during the day. Consolidated Timetable

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Characteristics of Time table management system

Features of timetable management software

1. Canceled timetable and substitutions: Ease of viewing canceled sessions and arranging substitute classes for those slots.
2. Subject electives with staff and different check-in locations: Assign subjects freely to teachers and monitor their

attendance through real-time check-in.

3. Staff's availability during a particular hour: View real-time the number of teachers available to fill in a class urgently.

4. Holistic timetable: Provides teachers with sufficient workload and gaps over the whole semester and boosts work productivity.

Benefits of the timetable management system for educational institutions

It allows classrooms to run smoothly on a given day.

It offers an error-free solution for classes to be scheduled systematically.

All subjects are given their due importance according to the syllabus.

It is secure since only those with relevant credentials can access the module and make changes.

It is a customizable solution. Any changes could be made without any conclusions or delays.

Feasibility Study of Timetable Management System

Feasibility Study

Before starting any project, it is always a good idea to conduct a feasibility study on the intended system. This is done to ensure that the net benefits are justifiable in comparison to overall costs and technical possibilities. As a result, a feasibility study was conducted to determine whether the intended system is feasible.

Technical Feasibility

The proposed Timetable Management System will be completely web-based and will require the use of available web technologies. UNIVOTEC makes a decision to use Python as the primary development language, with MySQL serving as the system's database management system. python and MySQL are both available for free use as open-source software licenses that do not require registration or purchase. The Apache web server is a free and reliable web server technology that will be used as a web server in the proposed system. All the above web technologies (Python, MySQL, Apache) are mature enough for enterprise-level software development as all of them are appearing in the software development industry for more than a decade and most of the developers having a good understanding of the technology. timetable: Provides teachers with sufficient workload and gaps over the whole semester and boosts work productivity.

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Operational Feasibility

Main stockholders of the system will be academic staff, executive staff, other officer level staff and undergraduate students. Therefore, no need to provide deep training and every one of the above categories knows how a web-based system to operates. UNIVOTEC already having dedicated staff to provide technical support as required. They have a very good understanding of web-based software since some other web based system currently in operation

available in the university. Web Based Timetable Management System for University of Vocational Technology (UNIVOTEC) 6 The available modern LAN system has been developed with web-based system operation in mind. Therefore, it is providing all required service such as security, availability, maintainability, robustness and backup for the system, and it will be the advantage of the system.

Economic Feasibility

The system gives added benefits, such as virtual administration and etc. at the successful implementation. Moreover, the system is built using highly reliable and freely available software products (General Public License) such as Python, MySQL and Apache. The use of available IT infrastructure (hardware, network, etc.) and the ability to virtually work from home would be an added advantage as a means of teleworking in the future. Taking things together, the intended system would contribute to enhancing the quality of the organization in a much cost-effective manner.

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Related Technologies

UVTTMS should be accessing all users, and therefore web application is much suitable for that. Users access

system via a uniform environment. It is critical to select the appropriate hardware, software, and technology while developing a better web-based system. The technology chosen as the development tools for the UVTTMS is explained in more detail below.

Web Application

This is a web application. To create an application, a combination of server-side script and client-side script (PYTHON, HTML, JavaScript, CSS, and so on) is commonly used.

Database Access

This is a real database that can store, update, delete and retrieve data. "UVTTMS store all the venues, lectures, subjects, students, and preferred timeslots details. UVTTMS always stores more than five years back data to analyze data. MySQL is used as my database. MySQL is the database of choice for the most demanding Web, E-commerce, SaaS, and Online Transaction Processing (OLTP) applications. It is a transaction-safe, ACID-compliant database that supports full commit, rollback, crash recovery, and row-level locking. MySQL provides the usability

scalability, and performance required by Facebook, Google, Twitter, Uber, and Booking.com"

GOALS OF TIMETABLE MANAGEMENT SYSTEM

Modules of the System

The module is a different segment of the system that processes a specific task. This facilitates system development and makes it more user-friendly.

As per requirements, the system mainly contains seven modules. Such as Login modules, Registration module, Lecture module, Course module, Subject module, Year plan module, Resources module, timetable generating module and Report module.

Login Module:

This will assist the user in logging into the system with their username and password. Only a user with a valid username and password can access their accounts.

It will facilitate in the authentication of the user who logs into the system. This prevents any anonymous user from entering the system and mishandling the records. It is superior to the manual method because it has no security measures governing who can and cannot access the system.

Timetable generating Module

This is the module generating timetable according to the resources available and with fulfill of institutes constrains. In this module, the user can generate a timetable with few options. University uses maximum resources in weekend, and it is essential to generate timetable with full optimization in weekend courses. Other hand, weekday programmes currently have enough resources with a minimum number of weekday degree programmes.

Therefore, UVTTMS has a facility to generate the same timetable with a defined set of constraints and export better one as user's satisfaction.

Subject module:

This module allows the admin to enter a new subject to the system. After submitting the subject each subject is given a "Subject ID Name" which is used to search the subject in the search box.

Lecture module:

This module allowed to add a new lecture to the system. The admin need to fill the details of the lecture. Each lecture is given a "Lecture ID Name" when its stored in the database. This Id name can be used by the admin to search lectures stored by entering the Id name in the search box.

Course module:

The administrator can create a new course, which is saved in the system's database. In order to accomplish this, this admin should enter the course name, the respective department of the course, maximum number of the student in the course and the section. After submitting the course, it is given an auto increment number (course code). This number is used to store the course in the database.

Resource module:

This module which given the details (Classroom Name, Location, Student Capacity and method) of each class room and labs. If a classroom is not added the admin can reenter the "Add resources Page" by clicking the "Add Classroom" button.

Report module:

Report module is used to get the details of time table. This module contains two parts.

Such as,

Semester time table report Year plan report

SELECTED SOFTWARE

Python is an interpreted, object-oriented, high-level programming language with dynamic semantics developed by Guido van Rossum. It was originally released in 1991.

Designed to be easy as well as fun, the name "Python" is a nod to the British comedy group Monty Python. Python is a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation. Python is dynamically typed and garbage-collected.

It supports multiple programming paradigms, including structured, object-oriented and functional programming.

Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together.

Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and packages, which encourages program modularity and code reuse.

The Python interpreter and the extensive standard library are available in source or binary form without charge for all major platforms, and can be freely distributed.

Often, programmers fall in love with Python because of the increased productivity it provides. Since there is no compilation step, the edit-test-debug cycle is incredibly fast. Debugging Python programs is easy: a bug or bad input will never cause a segmentation fault.

Instead, when the interpreter discovers an error, it raises an exception. When the program doesn't catch the exception, the interpreter prints a stack trace.

A source level debugger allows inspection of local and global variables, evaluation of arbitrary expressions, setting breakpoints, stepping through the code a line at a time, and so on.

The debugger is written in Python itself, testifying to Python's introspective power. On the other hand, often the quickest way to debug a program is to add a few print statements to the source:

the fast edit-test-debug cycle makes this simple approach very effective.

BULLET Education Suit-Automatic Timetabling Software

Bullet Education Suite timetable software offers higher education institutions a suite of software solutions designed to simplify the timetabling process. It comes with seven modules, as indicated below.

- **BULLET Timetable Education**

An automatic timetabling software capable of generating highly optimized timetables based on the interests of educational institutions. It enables you to make the most of your resources while also ensuring the satisfaction of your students and academic staff.

- **BULLET Calendar**

powerful space management solution with a decision support system designed to assist your Institution's daily operations, ensuring that you will not make any more management mistakes.

- **BULLET Info board**

one-of-a-kind university digital signage solution that will allow you to easily communicate with your academic community.

- BULLET Business Intelligence

Get real-time data on your institution's current overall performance. It enables massive resource optimization and significant cost savings.

- BULLET Interactive

An interactive web academic portal designed to provide academic staff and students with access to all of the information and materials they require for their daily activities.

- BULLET Exam Planer

With an automatic exam scheduling tool, you can easily accommodate all academic staff and students to their related activities, ensuring a perfect balance between students, supervisors, and classroom distribution.

Classroom Allocator

An automatic classroom management solution designed to assist your institution in making the most

of available classrooms in a cost-effective manner, ensuring maximum classroom utilization (Bullet Education , n.d.).

Furthermore, the Bullet timetable solution includes numerous useful features such as fully automatic timetable generation, assisted manual adjustments, error-free timetabling, comprehensive incoherency reports, multiple optimization methods, simple online timetable publication, and full data import and export.

“UniTime is a comprehensive educational scheduling system that allows for the creation of course and exam timetables, the management of changes to these timetables, the sharing of rooms with other events, and the task of students to individual classes. It is a distributed system that enables multiple universities and departmental schedule managers to collaborate on developing and modifying a schedule that meets their diverse organizational needs while minimizing student course conflicts. It can be used on its own to create and maintain a school's class and/or exam schedule, or it can be integrated with an existing student information system” (UniTime, n.d.).

The UniTime timetabling system comes with four main modules. It is

Course Scheduling and Management

Scheduling of Examinations

Event Management

Scheduling for Students

Course Scheduling

The primary goal of course scheduling is to place each course at a time (or set of times) that does not conflict with the time(s) assigned to any other required course by the students enrolled in it. This is relatively simple if only a few courses are taken in tandem with the course of interest. It becomes significantly more difficult as the number of courses requiring different time placements grows. The availability of faculty, rooms, and a variety of other constraints complicates matters even further.

Actual student course demands, an established set of curricula, historical patterns, or a combination of these data can be used to minimize student course conflicts. To arrive to a solution, student conflicts are weighed against faculty time and room preferences, as well as other desired interactions between classes.

UniTime incorporates the most recent research on solution algorithms designed to solve complex timetabling problems efficiently. It also has a fairly detailed course structure model that makes it simple to specify the links between course components with various styles of instruction (e.g., lecture, discussion, laboratory) and varying meeting time needs.

Furthermore, UniTime offers centralized, distributed, and hybrid approaches to creating a class schedule based on your needs. The course scheduling module can also be used to test different scenarios, such as the impact of fewer classrooms or changes in course requirements.

Course Management

“Even the best timetable may necessitate changes due to changing needs or resource availability. A change in a room or time assignment will be necessitated by an increase in demand for a course, the departure of a faculty member, or the loss of a classroom. UniTime enables users to easily search for alternatives that have the least impact on the overall

timetable, change the class schedule, and communicate these changes to affected students and other systems.

After the timetable is published, class adjustments are made using an interactive solver that generates a small selection of alternative time and/or room assignments for classes that will allow the proposed change to happen. Initially, the solver looks for alternative assignment adjustments that best meet all of the class requirements and preferences while not altering the assignment of more than two extra classes. The user can then choose the set of changes that best suits his or her needs from the options presented. Each choice includes details on any more student disagreements that may have arisen, as well as any other preferences that may have been violated. If a solution that affects only two other classes isn't possible, the search can be tweaked to accommodate more changes. Changes can be limited to allowing only room swaps, only time swaps, or both room and time swaps" (UniTime, n.d.).

Scheduling of Examinations

Each term, UniTime creates a comprehensive exam schedule that minimizes the amount of conflicting

exam placements for all students. It can also reduce the amount of back-to-back tests or students who have more than a certain number of exams in a day. This is especially beneficial for schools and institutions with often changing class offerings or a large number of multi-section courses that do not fit well into mapped exam timetables.

UniTime can be used to generate schedules for both midterm and final exams.

Scheduling for Students

“The student scheduling procedure is matching each student's set of needed classes to available class seats in order to meet all (or as many as possible) of the student's educational requirements. Individual student class time preferences may also exacerbate the situation. The most direct way to ensure that every student is able to attend all of their required courses is to create a class schedule after collecting all student course requirements. UniTime can be used to create a demand-based schedule and maximize the number of students who take the required courses. However, because this is not always possible, it is preferable to accommodate as many students' demands as possible with an existing schedule.

The major challenge of scheduling individual students to a pre-existing timetable in real time is ensuring that the decisions made by the initial students in the process do not unduly prevent later students from attending all of their necessary courses. When numerous portions of a course are given at various times, this becomes a problem. If all courses are offered at the same time and students are unable to attend the courses they require, a more flexible course schedule is required" (UniTime, n.d.).

Functional Requirements

These are the requirements that the end user specifically demands as basic facilities that the system should offer. All these functionalities need to be necessarily incorporated into the system as a part of the contract. These are represented or stated in the form of input to be given to the system, the operation performed and the output expected. They are basically the requirements stated by the user which one can see directly in the final product, unlike the non-functional requirements (Sandeep Jain, n.d.).

The following major components and their functions were recognized as the main business processes of the intended system.

- System Home page

- ✦ System home page display only User Login form and help option.
- ✦ System identifies correct user group type and direct to appropriate pages with given user privileges.
- User Login
 - ✦ System has three user login categories such as Admin, Lecturer and Student.
 - ✦ All Students of the UNIVOTEC have a user login.
 - ✦ Login user name will be same as student registration number.
 - ✦ Default password will be e-mail address and students able to change their password whenever their need.
 - ✦ If „Incorrect User name or Password“ message appears, then user need contact TSU Admin to reset password.
 - ✦ There is an accurate validate for user name and password.
 - ✦ Successful user login will show a welcome message and direct to employee home page based on their user category.

- Students home page
 - ✦ View the entire menu which students accessible for the category, other menu options should be disabled or not view.
 - ✦ Students Home page display timetable related to the student with resources allocated with lecturer notifications.
 - ✦ Student can change password view and print timetable and make a request through the system.
- Admin home page
 - ✦ This menu provides the all administration task.
 - ✦ Register all users to the system
 - ✦ Adding, editing, deleting entries in the master timetable.
 - ✦ Handle bulk printing from the system.
 - ✦ Admin can grant privileges to the other users as necessary.

Non-Functional Requirements

“Non-functional requirements may be more critical than the functional requirements. If these are not met, the system is useless” (Sommerville, 2004).

- Shall be platform-independent, i.e. shall run on any platform (Windows, Linux, Mac, etc.)
- Shall maintain high security and highly reliable
- It must be compatible with any other system
- Except for the authorized user/user group, no personal information about external stakeholders (students/affiliated bodies) shall be disclosed to system operators or other users
- The system shall respond quickly (0.2 seconds or below) to user actions
- Searching and sorting larger databases shall be comparatively less time consuming
- Shall provide a Graphical User Interface (GUI) for all end-users of the system
- The system shall be easy to learn (how to work with) and be able to learn by example
- Shall maintain a help facility for “how to use”

Shall be able to back up the entire system into a different driver/location

User Requirements

- The only end-user requirement is to have basic literacy of computer use, especially the techniques used in browsing the web.

System Requirements

- The system architecture is such that it conforms to 3-Tier Architecture having;
- a Backend in the bottom layer to store all data used in the system,
- an Application Server (Web Server) in the middle layer to process requests from and result back to the user
- a front end (any standard web browser) provides an interface to the end-user to deal with the system.

The Backend (Database Server)

- The database server is preferably a 2 CPU cores (recommended), 32GB or higher RAM,
- 1TB or higher storage space depending on the institution's data storage requirements. The backend is implemented with MySQL® 5.0 database management system.

The Application Server (Web Server)

It is observed that the performance of the webserver is one of the most vital factors pertaining to the speed of the system. Therefore, 2 CPU cores

(recommended), 32GB or higher RAM, 1TB or higher storage or more free Hard disk capacity, and access to Information Superhighway at a rate of 1 Mbps is recommended. It may run on Microsoft® Windows platform and Apache® 2.4 as the webserver software. PYTHON Hypertext Preprocessor (PYTHON) 7.1 is used as the scripting language.

Client

All that is needed by the client side is a computer that can run any standard web browser (preferably Google® Chrome) and access to Information Superhighway at a minimum rate of 1Mbps. Nevertheless, the more the performances of the computer and the network bandwidth, the faster the system processes as it would be.

Existing Similar Systems

A significant amount of effort and time was expended in finding other similar systems and approaches in order to study their benefits and drawbacks in order to implement the system in an effective and efficient manner.

TMS (Timetable Management System) provides a centralized, accurate view of all timetable and resource allocation activities. Technical Services department, as well as increasing department efficiency through standardization

of timetable and resource allocation processes. The Technical Services department is critical to the smooth running of the organization because it tracks and analyzes the time keeping and work patterns of academic activities. There are numerous applications available to assist the Timetable and Resource Allocation Department with their tasks, allowing for the automation of certain tasks and assisting in the organization of many other tasks.

UNITIME- Timetable System

“UniTime is a comprehensive educational scheduling system that allows for the creation of course and exam timetables, the management of changes to these timetables, the sharing of rooms with other events, and the task of students to individual classes.

It is a distributed system that enables multiple universities and departmental schedule managers to collaborate on developing and modifying a schedule that meets their diverse organizational needs while minimizing student course conflicts. It can be used on its own to create and maintain a school's class and/or exam schedule, or it can be integrated with an existing student information system” (UniTime, n.d.).

The UniTime timetabling system comes with four main modules. It is

- Course Scheduling and Management
- Scheduling of Examinations
- Event Management
- Scheduling for Students

Course Scheduling

The primary goal of course scheduling is to place each course at a time (or set of times) that does not conflict with the time(s) assigned to any other required course by the students enrolled in it. This is relatively simple if only a few courses are taken in tandem with the course of interest. It becomes significantly more difficult as the number of courses requiring different time placements grows. The availability of faculty, rooms, and a variety of other constraints complicates matters even further.

Actual student course demands, an established set of curricula, historical patterns, or a combination of these data can be used to minimize student course conflicts. To arrive to a solution, student conflicts are weighed against faculty time and room preferences, as well as other desired interactions between classes.

UniTime incorporates the most recent research on solution algorithms designed to solve complex timetabling problems efficiently. It also has a fairly detailed course structure model that makes it simple to specify the links between course components with various styles of instruction (e.g., lecture, discussion, laboratory) and varying meeting time needs.

Furthermore, UniTime offers centralized, distributed, and hybrid approaches to creating a class schedule based on your needs. The course scheduling module can also be used to test different scenarios, such as the impact of fewer classrooms or changes in course requirements.

Course Management

“Even the best timetable may necessitate changes due to changing needs or resource availability. A change in a room or time assignment will be necessitated by an increase in demand for a course, the departure of a faculty member, or the loss of a classroom. UniTime enables users to easily search for alternatives that have the least impact on the overall timetable, change the class schedule, and communicate these changes to affected students and other systems.

After the timetable is published, class adjustments are made using an interactive solver that generates a small selection of alternative time and/or room

assignments for classes that will allow the proposed change to happen. Initially, the solver looks for alternative assignment adjustments that best meet all of the class requirements and preferences while not altering the assignment of more than two extra classes. The user can then choose the set of changes that best suits his or her needs from the options presented. Each choice includes details on any more student disagreements that may have arisen, as well as any other preferences that may have been violated. If a solution that affects only two other classes isn't possible, the search can be tweaked to accommodate more changes. Changes can be limited to allowing only room swaps, only time swaps, or both room and time swaps" (UniTime, n.d.).

Scheduling of Examinations

Each term, UniTime creates a comprehensive exam schedule that minimizes the amount of conflicting exam placements for all students. It can also reduce the amount of back-to-back tests or students who have more than a certain number of exams in a day. This is especially beneficial for schools and institutions with often changing class offerings or a large number of multi-section courses that do not fit well into mapped exam timetables.

UniTime can be used to generate schedules for both midterm and final exams.

Scheduling for Students

"The student scheduling procedure is matching each student's set of needed classes to available class seats in order to meet all (or as many as possible) of the student's educational requirements. Individual student class time preferences may also exacerbate the situation. The most direct way to ensure that every student is able to attend all of their required courses is to create a class schedule after collecting all student course requirements. UniTime can be used to create a demand-based schedule and maximize the number of students who take the required courses. However, because this is not always possible, it is preferable to accommodate as many students' demands as possible with an existing schedule.

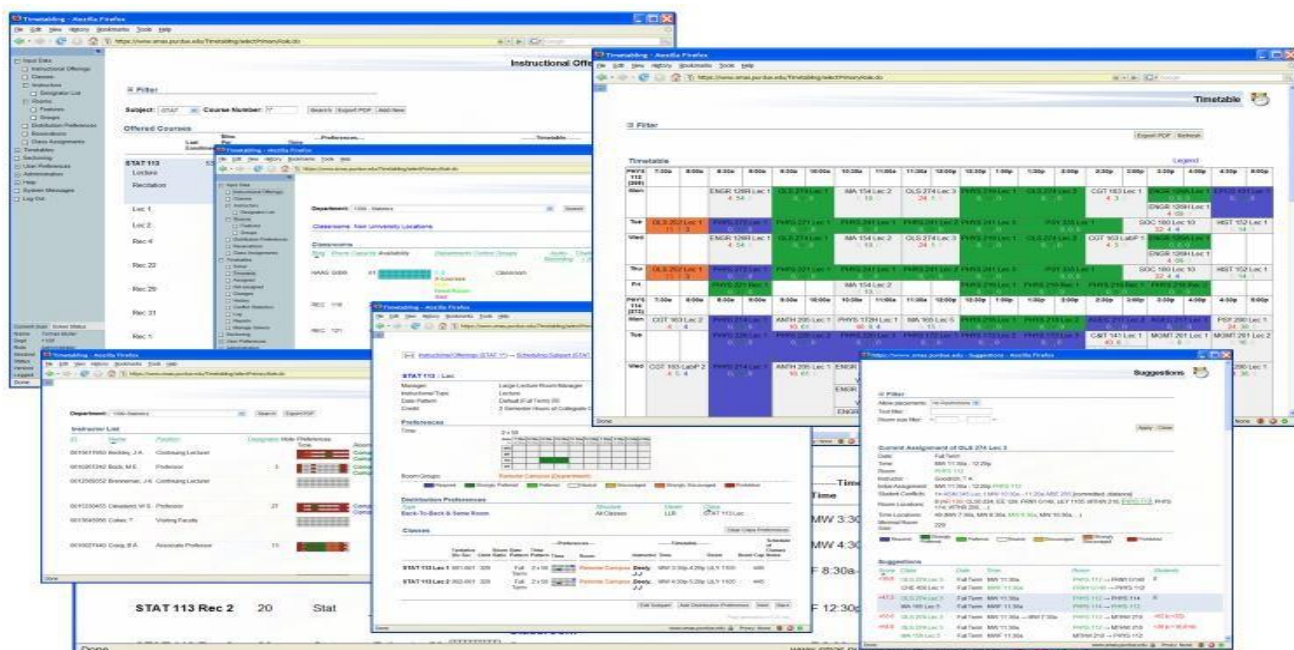
The major challenge of scheduling individual students to a pre-existing timetable in real time is ensuring that the decisions made by the initial students in the process do not unduly prevent later students from attending all of their necessary courses. When numerous portions of a course are given at various times, this becomes a problem. If all courses are offered at the same time and students are unable to attend the courses they require, a more flexible course schedule is required" (UniTime, n.d.).

Figure shows timetable created by UniTime Timetable Management System.

Figure .Timetable created with UniTime Timetable Management System

EduSys Auto Timetable System

“The EduSys Web & Cloud-based System offers powerful Classroom Timetable Scheduling Software with full class, course, and exam scheduling capabilities for school, college, and university administrators and instructors to design, administer, and alter classroom timetables and curricula. The



online timetabler software eliminates paper dependency by providing secure, automated, and digitized solutions for coordinating examination schedules, managing exam rooms/centers/venue, archiving important events and dates, allocating

students to courses, planning class timings for teachers and students, and much more" (Anon., 2017).

Edusys system contains the following main modules with a set of valuable features.

Auto Timetable Schedule

The Auto timetable schedule function automates the scheduling of classes, exams, and courses for students, teachers, and various classrooms by considering all potential "What-if" scenarios and results. Furthermore, the timetabler software includes usercentric and easy-to-use tools that allow educators to view, organize, and generate timetables for each teacher/class/grade, tailor timetables, create and pin to-do lists, schedule substitute replacements for absent staff, manage and organize events on the calendar, and much more on a smartphone, tablet, and computer.

Advance Allocation Tool

The Timetable Software includes sophisticated digital allocation methods for assigning students to courses and classes, assigning instructors to classrooms, and distributing replacement professors online. Furthermore, the system allows administrators to assign and change timetables for specific teachers and pupils, alleviating the burden of manually

preparing class schedules and distributing them to each person.

Attendance Management

The Timetable Software works in tandem with all RFID and biometric attendance control systems. The software allows educational institutions such as schools, colleges, and universities to track and record student and instructor attendance, send SMS/Email messages to parents/guardians alerting them of their children's attendance, monitor staff and faculty activities and whereabouts, and much more.

Print & Share

With a single click, you can print and share, personalized, and individual timetable schedules in bulk with colleagues, students, and parents. By utilizing a mobile app on any smart device or the Web version on any computer, you may review and export the reports and schedules to PDF, Word, and many other formats online.

Summary of Similar Systems

In order to execute the system in an effective and efficient manner, a significant amount of work and time was spent in finding other comparable systems and methods and studying their positive and negative elements, as mentioned above. The following are the features of the available system. Some

aspects of the existing systems have been enhanced, while others are important to the proposed system.

Database Design:

TEACHERS

TEACHER_ID
TEACHER_NAME
WEEK HOUR

SUBJECT

SUBJECT_ID
SUBJECT_NAME

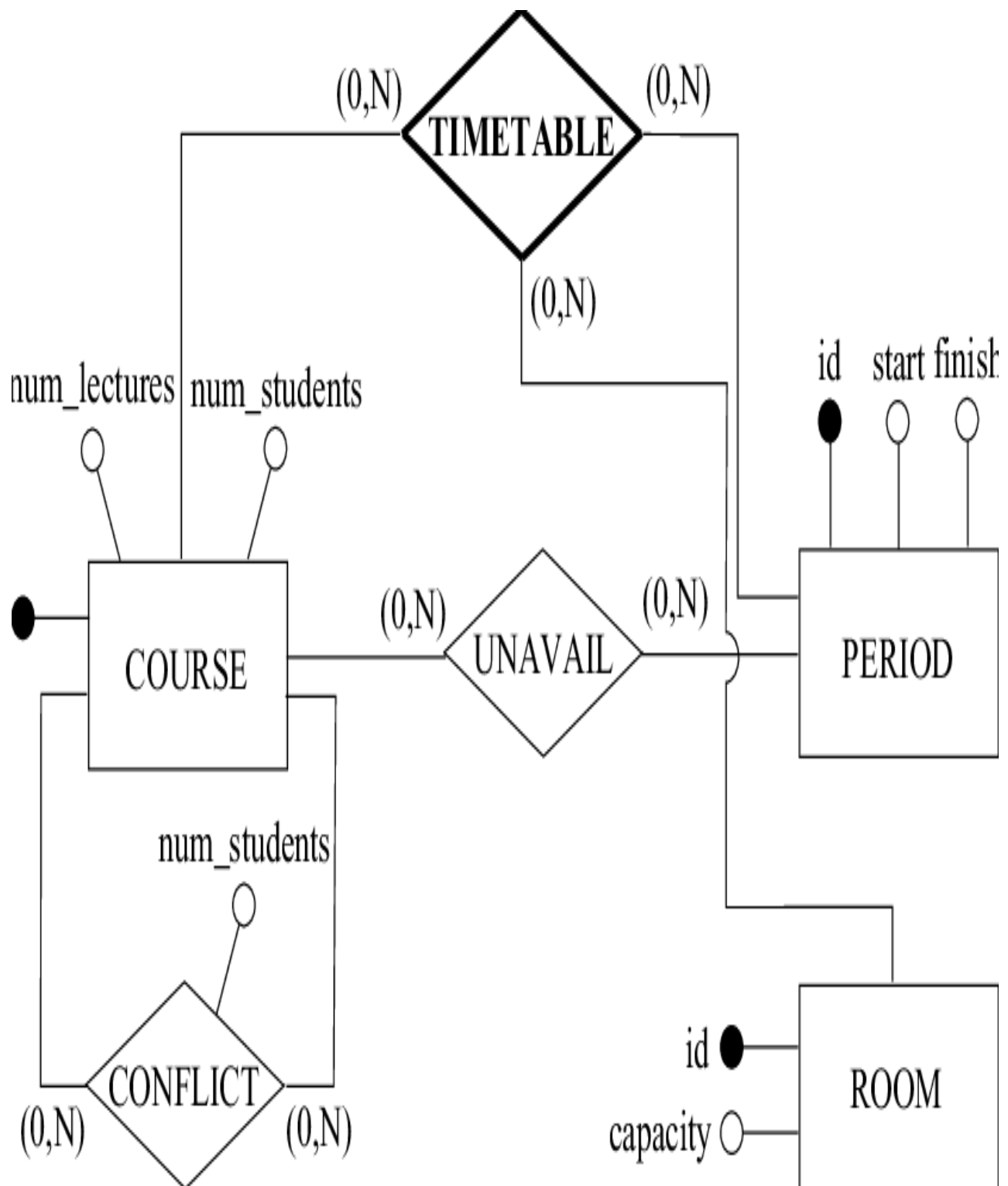
TEACHER_SUBJECT
TEACHER_ID
SUBJECT_ID

CLASSES

CLASS_ID
CLASS_NAME

CLASS_TEACHER
CLASS_ID
TEACHER_ID
TEACHER_HOUR
MAX_HOUR

ER DIAGRAM



CLASS DIAGRAM

Class diagrams are an essential component of object-oriented analysis and design. Class diagrams depict the system's classes, their interrelationships (including inheritance, aggregation, and association), and the classes' activities and characteristics.

Class diagrams are used for a wide variety of purposes, including both conceptual/domain modeling and detailed design modeling.

The class diagram below Figure depicts the idea. With a single click, you can print and share, personalized, and individual timetable schedules in bulk with colleagues, students, and parents. By utilizing a mobile app on any smart device or the Web version

Researcher: Dr. Jyoti Wansontale and Pooja Vaidya

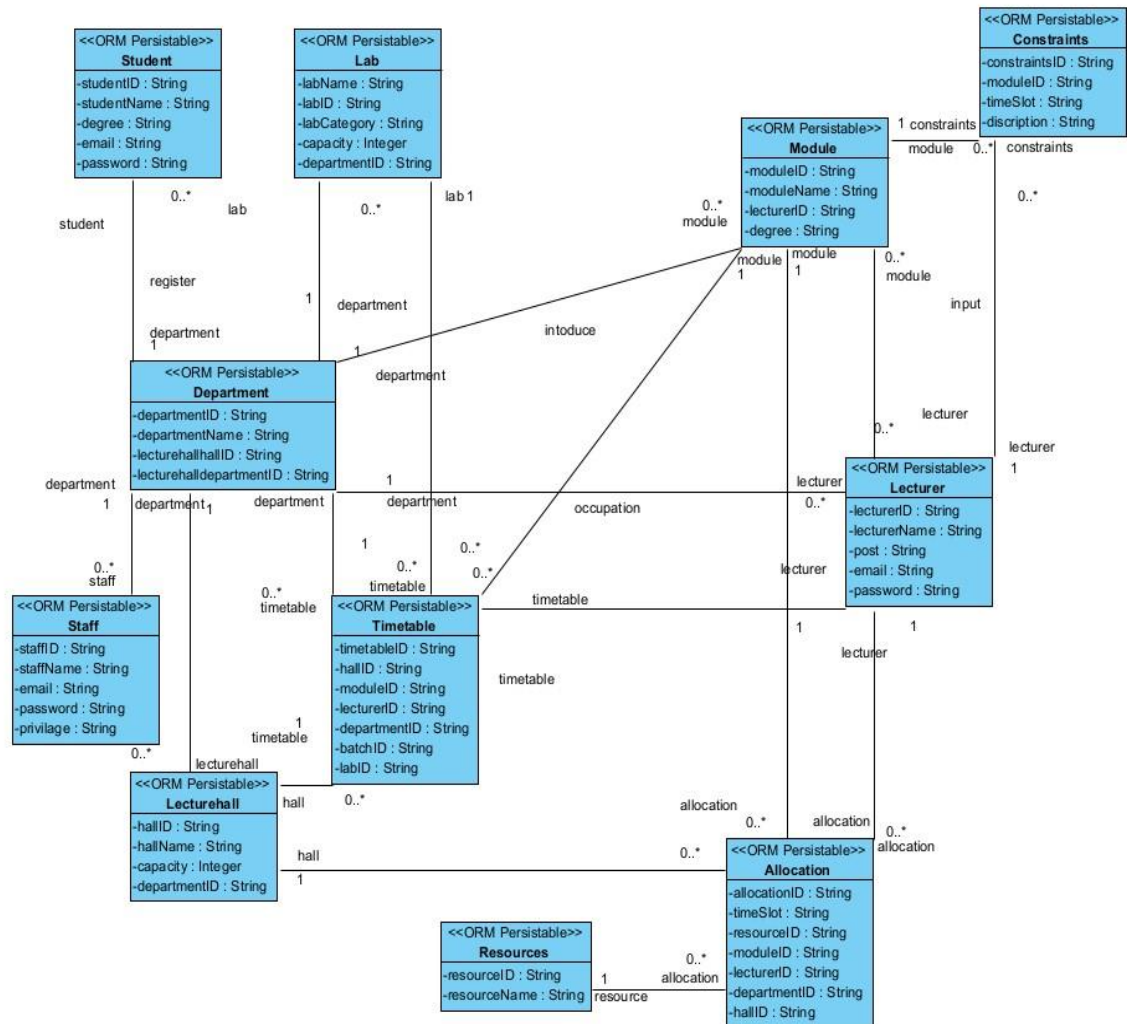


Figure. Class diagram of the system

Activity Diagram

Activity diagrams show how a business process or a use case's activities flow in a logical order. They may also be used to simulate the activities that will be taken when an operation is carried out, as well as the outcomes of those actions. It demonstrates how actions in the system are interdependent in the diagram below in figure. International Journal of Educational Research and Information Science 2021; 7(1): 1-13 7 Departments, Dean and the Examination Committee. The service providers for the proposed Web-based Timetable is the System Developer. The governing bodies for the system include the Board of Governors for the University, Academic Board and the Student Representative Council. Figure 2. Use Case Diagram for the Web-based Timetable System. 4.2. Activity Diagram Unified Modeling Language like activity diagrams is purposely designed to model out both computational and organizational process [13]. The essence of Activity Diagram for the proposed Web-based Timetable System is to model the workflow of its Use Case to show paths within it. It will show the sequence of an activity from the other. The Activity Diagram will illustrate existing functionalities and how those functionalities coordinate with other pieces

functionalities of the system. The general overview or description of the for the Timetable System activity diagram is depicted in figure

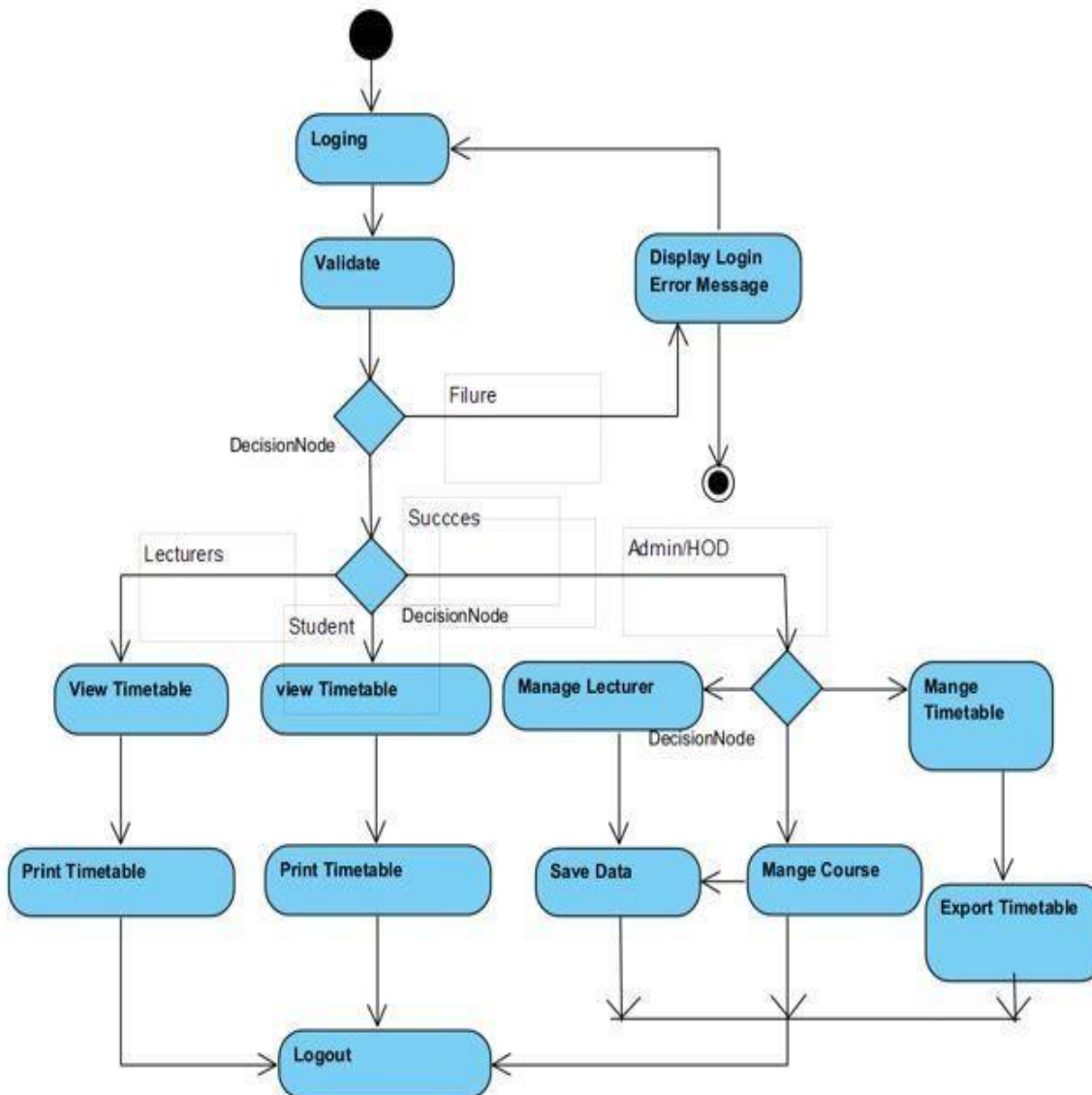


Figure .Activity Diagram of the System

Sequence Diagram

During the execution of a use case or any action, this defines how objects communicate with one another via messages. They show how messages are delivered and received between objects, as well as the order in which messages are sent. It also describes how operations are carried out according to the time of operation. Figure 3.4 depicts the Timetable Management System sequence diagram.

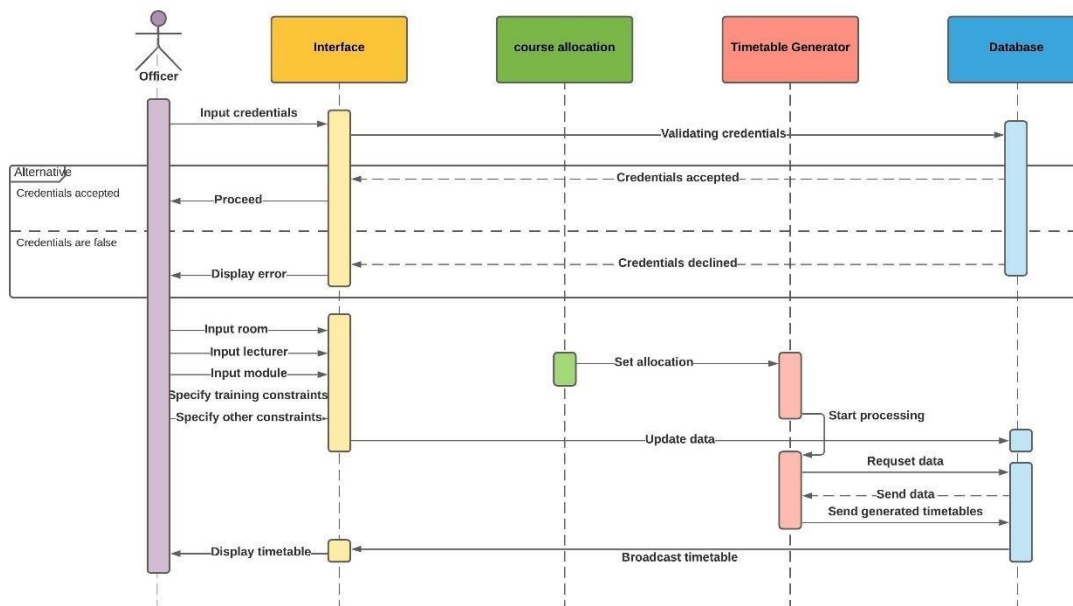


Figure .Sequence diagram of the System

Database Design

The database is an essential component of every software system. Defining the database entails describing the data types, structures, and restrictions for the data that will be stored in it. Entity-Relationship (ER) diagrams, ER Mapping are used to illustrate the design of database.

The Entity Relationship (ER) Model

Figure 3.5 depicts the Entity-Relationship diagram of the timetable management system's major data elements and their relationships.

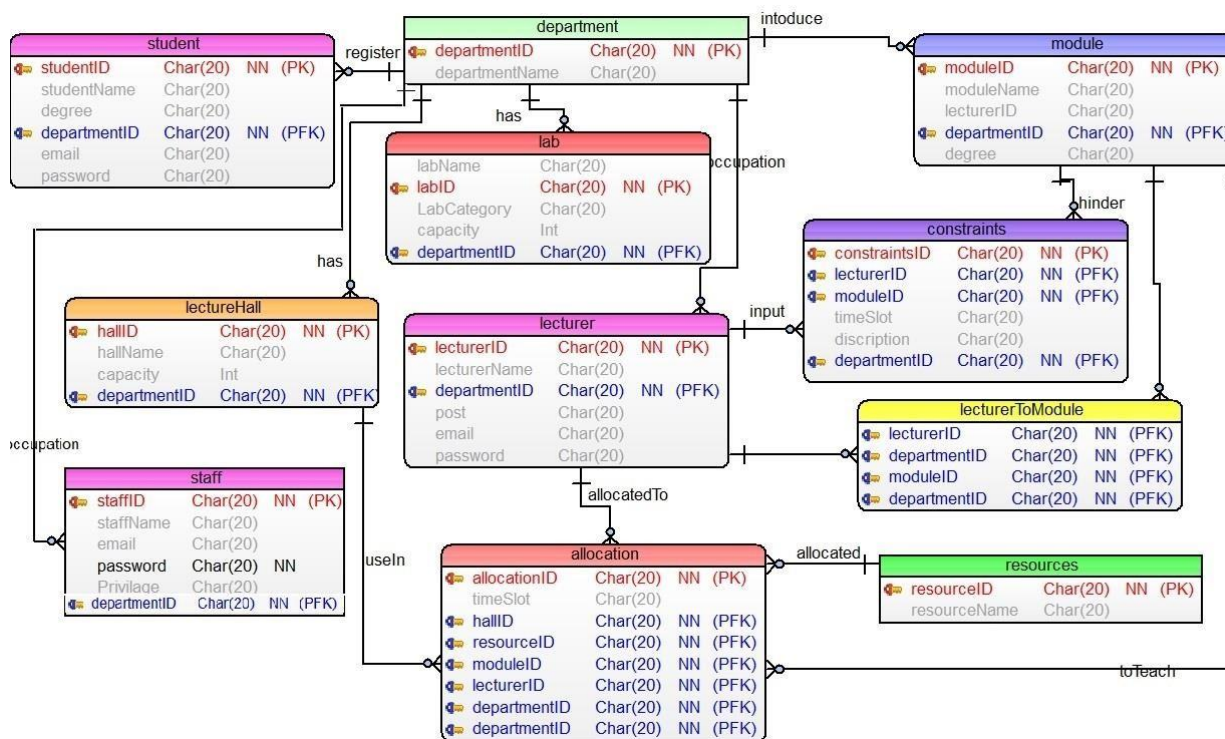


Figure .The Entity-Relationship diagram**User Interface Design**

The design of the user interface must be done with great attention since it will be critical to the system's success. The user interface was found to be simplistic, yet detailed enough to keep consumers directed around the site. This project's user interface design and development required a long time since it had to accommodate to the many stakeholders' demands, as detailed in the preceding chapters.

Somerville (Sommerville, n.d.) recommends the following key aspects when developing user interfaces.

1. Understand the business functions – The major business functions performed by different stakeholders identified in the Analysis phase have been put into careful consideration to provide the required functionality.

2. Apply general principals of good screen design as depicted by the figure below

Related Technologies

UVTTMS should be accessing all users, and therefore web application is much suitable for that. Users access system via a uniform environment. It is critical to select the appropriate hardware, software, and technology while developing a better web-based system. The technology chosen as the development tools for the UVTTMS is explained in more detail below.

Web Application

This is a web application. To create an application, a combination of server-side script and client-side script (PYTHON, HTML, JavaScript, CSS, and so on) is commonly used.

Database Access

This is a real database that can store, update, delete and retrieve data. "UVTTMS store all the venues, lectures,

subjects, students, and preferred timeslots details.

UVTMS always stores more than five years back data to analyze data. MySQL is used as my database. MySQL is the database of choice for the most demanding Web, E-commerce, SaaS, and Online Transaction Processing (OLTP) applications. It is a transaction-safe, ACID-compliant database that supports full commit, rollback, crash recovery, and row-level locking. MySQL provides the usability, scalability, and performance required by Facebook, Google, Twitter, Uber, and Booking.com” (MySQL, n.d.).

Related Design Strategies

“Given any project requirement, there are many alternative strategies that could be used in achieving the desired objectives/outcomes. However, the selection of a design strategy mainly depends on factors such as the complexity of the project/system, the budget of the system and the time constraints” (Bentley, 2007).

At the design phase of Timetable Management System, it was considered to evaluate the feasibility of alternate design strategies such as Stand-alone system, Mobile

application, Commercial Application Package strategy and Web based system.

Standalone system

A stand-alone system is a software application that runs without the need for any other software beyond the operating system. The majority of standalone systems are platform dependent. As a result, establishing the specific operational environment may be prohibitively costly. Because each terminal must be maintained separately, the deployment, upgrading, and maintenance operations take time. Because isolated systems are limited to a single physical place, their applicability is limited. There will be no exchange of information.

Mobile Application

“As of 2019, 3.2 billion individuals globally possessed a smartphone, which means that one out of every three mobile users can run applications on their devices. Combine this with the ongoing astounding surge in app use — downloads climbed from just over 20 billion in 2016 to well over 30 billion in the first quarter of 2020 – and it's obvious that they're a popular way for people to connect with their mobile devices” (Mews, n.d.).

Frequently increasing usage of smart phones, efficiency, performance and capability of run application as well as speed of mobile networks increase day by day.

Development of infrastructures of mobile device has create nice opportunity to develop mobile applications for the business.

The mobile App development having following advantages as well as disadvantages.

The Advantage

Always visible on the user's screen. As a result, your mobile app has a high probability of being displayed at all times.

More personalization and loyalty. The designers of an app can customize their customer's user experience.

A great tool of analytics. The app analytics provide the most popular content as well as other useful information.

Easy to interact with users. You can do so thru two types of messages existing in mobile apps - push and in-app notifications.

The content is available offline. With an app, It is now possible to download material and subsequently enjoy it

when flying or climbing in the mountains without access to Wi-Fi.

More optimized than website. An app is designed to work with a range of phone models, screen resolutions, and software.

Apps are more responsive than webpages. Apps may keep all data on the device, but mobile sites must connect to web servers.

Using device features. For example, you can easily use a camera or a GPS .

Providing extra revenue channels. For example, the freemium model provides basic software functions for free while charging for additional services. It also enables for the creation of several plans to meet the demands of different users.

The Disadvantage

The high cost of app development

The process of developing an app is complicated.

Long-term investment is required to guarantee compliance.

Technologies not mature enough

Knowledge of technology not easily available

Lack of clarity

The Commercial Application Package Implementation Strategy

These packages could be bought or obtained free (if it is Open Source) and customized according to the business requirements of the organization. These are also known as Commercial Off-the-Shelf Systems. Some advantages of these systems are.

Because substantial programming is not required, new systems may be installed more rapidly.

More suited for organization with less human resources for in-house system development

Subject to continuous improvement as the vendor distributes the system for use of anyone interested (as in the case of Open source software) Disadvantages

The successes of these systems depend on the vendor reputation and stability in the industry.

A purchased system may not cater the organizations requirement adequately. It may sometimes be more or less featured.

Alternate Solutions Evaluation

There are several alternative solutions available in software industry but required to select the correct tool for system based on the requirement gathered, functional and nonfunctional requirement. After successful completion of Feasibility study, Requirement Analysis and designing task will continue, and then need to find a solution to perform coding, Testing and Implementation tasks.

Before start coding need to decide whether to build windows based standalone application or web-based application using client server architecture. Since this a Timetable Management System all students of the university need access to the system. Based on the feasibility study done web-based application using client server architecture is selected because,

High Availability - If all employees need access to the system there should be a high availability, web-based system provide high availability then windows application.

Easy to maintain - During the software development life cycle maintain is the task which takes more time, in client server architecture maintains is very easy because

configurations and implementations need to be done only in the server, not to do each and every client computer.

Low cost - In client server only server need to be a high powerful computer, due to this it can be implemented with low cost.

Portability - Web based system is very easy to access from anywhere via network.

Centralization entails controlling access, resources, and data security through a server.

Scalability entails the capacity to update any part as needed.

Flexibility entails the ease with which new technologies may be implemented into the system.

Interoperability means that all components (clients, network, and servers) cooperate with one another.

Once the Architecture is selected then need to select the programming language, IDE and database to be used to develop the system. Since this is one-year project with limited time constrain therefore need to think about the language which developer familiar and which support for Rapid development technology with client satisfaction. Based on the detail analysis done selected PYTHON as

web technology with and selected MySQL as backend database.

Selected Solution Description and Justification

The system architecture for the proposed system is based on the popular 3-Tier Architecture used in most web-based systems. The User Interface Layer is used to provide the end-user a platform-independent interface to work with the system. It basically consists of the web browser and a firewall/proxy for enhanced security. The business Layer is where the entire business logic is defined and implemented. System processes are done in this layer. User requests are captured and handled in a way that the requests are sent to the database server (sometimes via Web Services when different databases are used) and queried for the desired output. PYTHON Hypertext Preprocessor is used to manipulate this layer. This layer consists of a web/application server.

The backend or the Data Layer stores administrates and manages the data used in the entire system. Query requests from the business layer are processed here. The popular MySQL® database management system is used here. The following diagram helps identify the idea of the architecture of this system. The system architecture for

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The backend or the Data Layer stores administrates and manages the data used in the entire system. Query requests from the business layer are processed here. The popular MySQL® database management system is used here. The following diagram helps identify the idea of the architecture of this system.

The concept is shown below in Figure

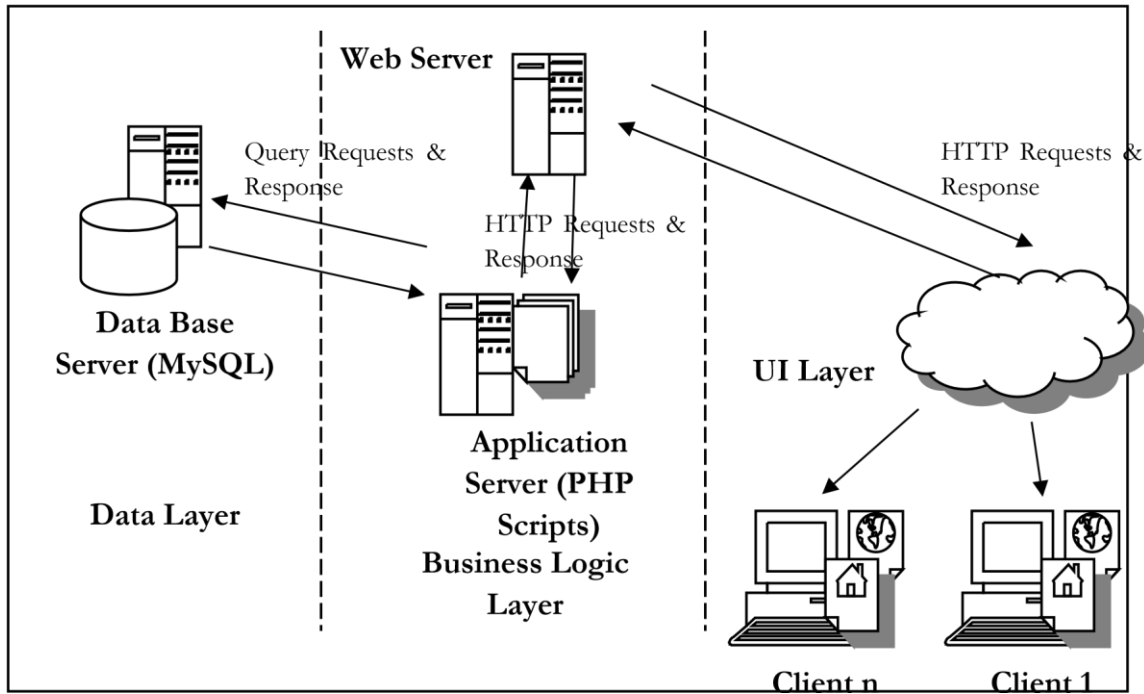


Figure .The Three-Layered Architecture

Alternate Solution Evaluation

"Given any project requirement, there are many alternative strategies that could be used in achieving the desired objectives/outcomes. However, the selection of a design strategy mainly depends on factors such as the complexity of the project/system, the budget of the system and the time constraints" (Bentley, 2007).

At the design phase of Timetable Management System, It was investigated to see if other solution techniques like Model-Driven Development, Rapid Application Development, and Commercial Application Packages were feasible.

The Model-Driven Development Strategy

This system development strategy stresses the creation of system models to aid in the visualization and analysis of problems, the definition of business requirements, and the design of information systems. The advantages of this strategy are;

- It is a more formal and structured design strategy.
- The requirements specification is more detailed and well-documented.
- Because diagrams are used to convey concepts, it is easier to validate business needs and system architectures.
- Alternative technological solutions are easier to discover, comprehend, and assess.
- System construction becomes easier and logical due to thorough (complete) analyze and design.

Disadvantages

- It takes time to collect information, create models, and evaluate them.
- Models could only be as good as the understanding of the users about the system

The full completion of one phase allows the developer to step into the other phase. Users must specify all requirements before design and the design must fully document technical specification before construction.

The Rapid Application Development (RAD) Strategy

This strategy supports fast development of the system thus saving time and money for the system developers.

- It is beneficial for projects with ambiguous and imprecise needs.
 - It involves active user and management participation to increase the end-user enthusiasm for the project
 - Errors and omissions are more likely to be identified in prototypes than in system models.
 - It is iterative and incremental and supports fast development of the project
- Disadvantages

- It is claimed that RAD promotes "a code implements and repair mentality" This raises the system's lifetime operating and maintenance expenses.
- RAD prototypes may dissuade system analysts from investigating more appropriate options.
- The emphasis on speed might have a negative influence on system quality.

The Commercial Application Package Implementation Strategy

These packages could be bought or obtained free (if it is Open Source) and customized according to the business requirements of the organization. These are also known as Commercial Off-the-Shelf Systems. Some advantages of these systems are;

- Because substantial programming is not required, new systems may be implemented more rapidly.
- More suited for organization with less human resources for in-house system development

- Subject to continuous improvement as the vendor distributes the system for use of anyone interested (as in the case of Open source software)
- Disadvantages
- The successes of these systems depend on the vendor reputation and stability in the industry.
- A purchased system may not cater the organizations requirement adequately. It may sometimes be more or less featured.

Design Strategy Selection

As it was observed in the previous sections, every design strategy has its own advantages and disadvantages based on the use. Therefore, a selection of a mixed (hybrid) strategy has been used throughout the system. It uses the formality and structured nature of Model-Driven Design and the rapidness and simplicity of the RAD strategy.

The reasons are being that the system is undertaken as an undergraduate project and hence must be developed in a more formal manner allowing any future developments to be made if required. However, it still needs to involve the end user actively throughout the project life cycle.

Moreover, the Object-Oriented System Development (OOSD) in combination with Agile/Extreme programming could also be used with RAD strategy. Another fact is that they support iterative and incremental development. As mentioned by (Scott W. Ambler, 2001), they help ease the design overhead by allowing the designer to model Just Barely Good Enough (JBGE) iterative versions of the software system.

“The use of Model-View-Controller (MVC) architecture and the Factory Design Pattern helps reduce code maintenance overheads. It eases up debugging and helps maintain the clarity of the code. Moreover, it enhances the code reusability and the independence between the model, view and the controller layers” (Ed Lecky-Thompson, 2004).

The design starts with Architectural Envisioning that depicts how different layers are collaboratively working together to make the system more secure, updatable and platform neutral.

As the phase moves on, it further concentrates on logical and physical design of data and processes and user interface design.

Use case Diagram

"The Use case diagram is used to identify the primary elements and processes that form the system. The primary elements are termed as "actors" and the processes are called "use cases".

The Use case diagram shows which actors interact with each use case. A use case diagram captures the functional aspects of a system.

More specifically, it captures the business processes carried out in the system. Also Use case diagrams define the requirements of the system being modeled and hence are used to write test scripts for the modeled system" (Anon., 2003). It eases up debugging and helps maintain the clarity of the code. Moreover, it enhances the code reusability and the independence between the model, view and the controller layers" (Ed Lecky-Thompson, 2004).

The design starts with Architectural Envisioning that depicts how different layers are collaboratively working together to make the system more secure, updatable and platform neutral.

As the phase moves on, it further concentrates on logical and physical design of data and processes and user interface design.

PROJECT TESTING

- Here various test plans and cases are described used in the testing of different modules and the overall system.

Test Plan

A test plan is a procedure used to schematically analyze and find the shortcomings or mistakes, if any, in a technical system or product.

Through experimental runs of the unit, this process is used to make sure that the unit can function to full capacity, both effectively and efficiently, once it is in full operation.

Therefore, the test plan clearly states how the unit or the unit or system will be tested, what aspects of it will be tested, who will perform the tests, where and for how long the testing will occur and the overall effectiveness of such testing.

Software Testing is an empirical investigation conducted to provide stakeholders with information about the quality

of the product or service under test, with respect to the context in which it is intended to operate. A test plan is a procedure used to schematically analyze and find the shortcomings or mistakes, if any, in a technical system or product.

Through experimental runs of the unit, this process is used to make sure that the unit can function to full capacity, both effectively and efficiently, once it is in full operation.

Therefore, the test plan clearly states how the unit or the unit or system will be tested, what aspects of it will be tested, who will perform the tests, where and for how long the testing will occur and the overall effectiveness of such testing.

Software Testing also provides an objective, independent view of the software to allow the business to appreciate and understand the risks at implementation of the software.

Test techniques include, but are not limited to, the process of executing a program or application with the intent of finding software bugs.

It can also be stated as the process of validating and verifying that a software program/application/product meets the business and technical requirements that guided its design and development, so that it works as expected and can be implemented with the same characteristics. Software Testing is performed for each and every possible test case in our entire project for checking the expected validations and correctness of flow of control from one page to another as requirement of user. Software Testing,

depending on the testing method employed, can be implemented at any time in the development process, however the most test effort is employed after the requirements have been defined and coding process has been complete.

Testing can never completely identify all the defects within software.

Instead, it furnishes a criticism or comparison that compares the state and behavior of the product against oracles² principles or mechanisms by which someone might recognize a problem. These oracles may include (but are not limited to) specifications, contracts, comparable Software Testing is an empirical investigation conducted to provide stakeholders with information about the quality of the product or service under test, with respect to the context in which it is intended to operate.

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CODE IMPLEMENTATION OF TIMETABLE MANAGEMENT SYSTEM

```
import tkinter as tk
from tkinter import ttk
from tkinter import messagebox
import os, sys
sys.path.insert(0, 'windows/')
import timetable_stud
import timetable_fac
import sqlite3

def challenge():
    conn = sqlite3.connect(r'files/timetable.db')

    user = str(combo1.get())
    if user == "Student":
```



```
cursor = conn.execute(f"SELECT PASSW,  
SECTION, NAME, ROLL FROM STUDENT WHERE  
SID='{id_entry.get()}'")
```

```
cursor = list(cursor)
```

```
if len(cursor) == 0:
```

```
    messagebox.showwarning('Bad id', 'No such  
user found!')
```

```
    elif passw_entry.get() != cursor[0][0]:
```

```
        messagebox.showerror('Bad pass', 'Incorret  
Password!')
```

```
    else:
```

```
        nw = tk.Tk()
```

```
        tk.Label(  
            nw,  
            text=f'{cursor[0][2]}\tSection:  
{cursor[0][1]}\tRoll No.: {cursor[0][3]}',  
            font=('Consolas', 12, 'italic'),
```

```
        ).pack()
        m.destroy()
        timetable_stud.student_tt_frame(nw,
cursor[0][1])
        nw.mainloop()

elif user == "Faculty":
    cursor = conn.execute(f"SELECT PASSW, INI,
NAME, EMAIL FROM FACULTY WHERE
FID='{id_entry.get()}'")
    cursor = list(cursor)

if len(cursor) == 0:
    messagebox.showwarning('Bad id', 'No such
user found!')

elif passw_entry.get() != cursor[0][0]:
    messagebox.showerror('Bad pass', 'Incorret
Password!')
```

else:

nw = tk.Tk()

tk.Label(

nw,

text=f'{cursor[0][2]} ({cursor[0][1]})\tEmail:

{cursor[0][3]}',

font=('Consolas', 12, 'italic'),

).pack()

m.destroy()

timetable_fac.fac_tt_frame(nw, cursor[0][1])

nw.mainloop()

elif user == "Admin":

if id_entry.get() == 'admin' and

passw_entry.get() == 'admin':

m.destroy()

os.system('py windows\\admin_screen.py')

```
# sys.exit()

else:

    messagebox.showerror('Bad Input', 'Incorret
Username/Password!')
```

```
m = tk.Tk()
```

```
m.geometry('400x430')
```

```
m.title('Welcome')
```

```
tk.Label(
```

```
    m,
```

```
    text='TIMETABLE MANAGEMENT SYSTEM',
```

```
    font=('Consolas', 20, 'bold'),
```

```
    wrap=400
```

```
).pack(pady=20)
```

```
tk.Label(  
    m,  
  
    text='Welcome!\nLogin to continue',  
    font=('Consolas', 12, 'italic')  
).pack(pady=10)
```

```
tk.Label(  
    m,  
    text='Username',  
    font=('Consolas', 15)  
).pack()
```

```
id_entry = tk.Entry(  
    m,  
    font=('Consolas', 12),  
    width=21
```

```
)
```

```
id_entry.pack()
```

```
# Label5
```

```
tk.Label(
```

```
    m,
```

```
    text='Password:',
```

```
    font=('Consolas', 15)
```

```
).pack()
```

```
# toggles between show/hide password
```

```
def show_passw():
```

```
    if passw_entry['show'] == "●":
```

```
        passw_entry['show'] = ""
```

```
        B1_show['text'] = '●'
```

```
        B1_show.update()
```

```
elif passw_entry['show'] == "":  
    passw_entry['show'] = "●"  
    B1_show['text'] = '○'  
    B1_show.update()  
    passw_entry.update()
```

```
pass_entry_f = tk.Frame()  
pass_entry_f.pack()
```

```
# Entry2
```

```
passw_entry = tk.Entry(  
    pass_entry_f,  
    font=('Consolas', 12),  
    width=15,  
    show="●"  
)  
passw_entry.pack(side=tk.LEFT)
```

```
B1_show = tk.Button(
    pass_entry_f,
    text='○',
    font=('Consolas', 12, 'bold'),
    command=show_passw,
    padx=5
)
B1_show.pack(side=tk.LEFT, padx=15)

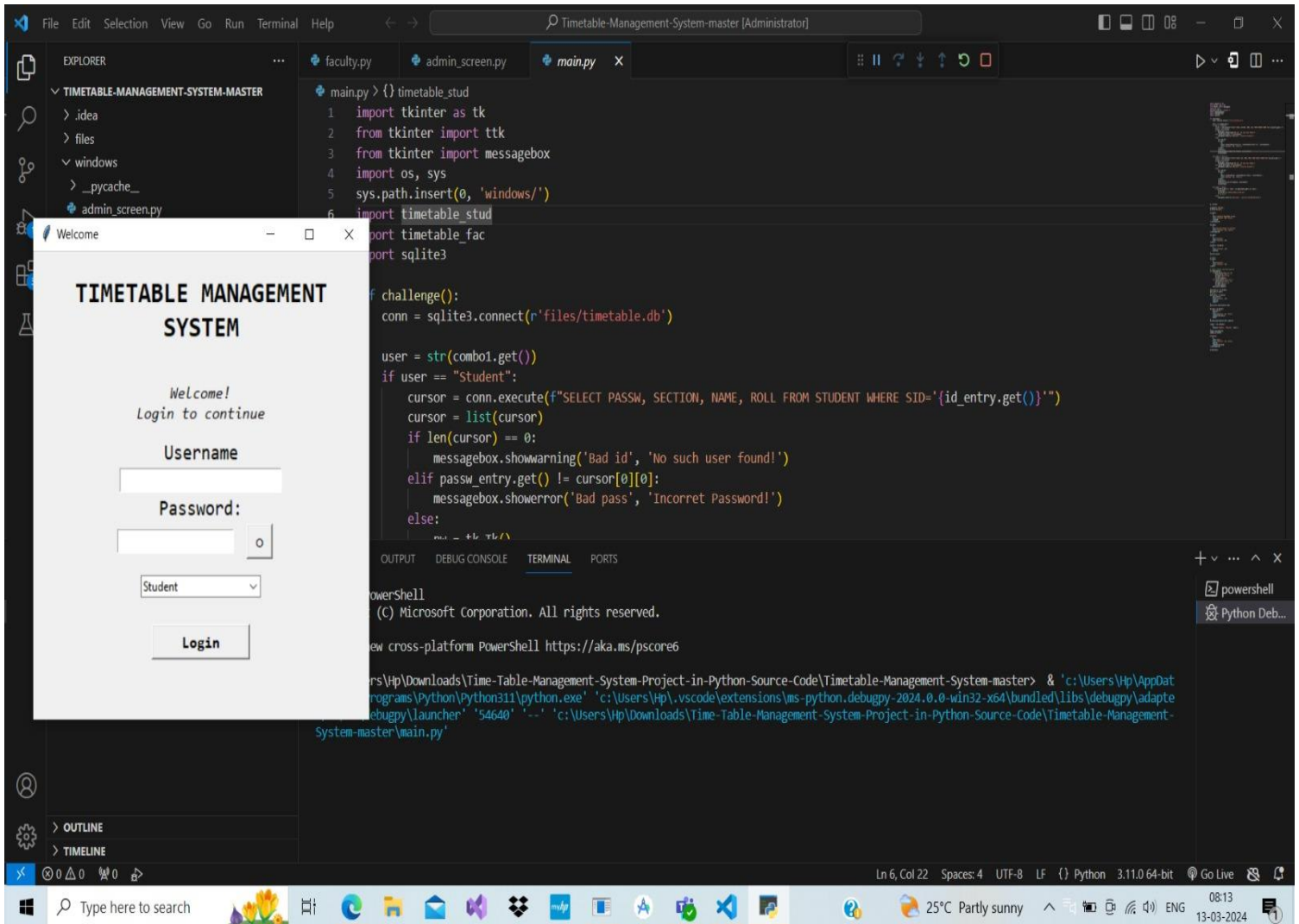
combo1 = ttk.Combobox(

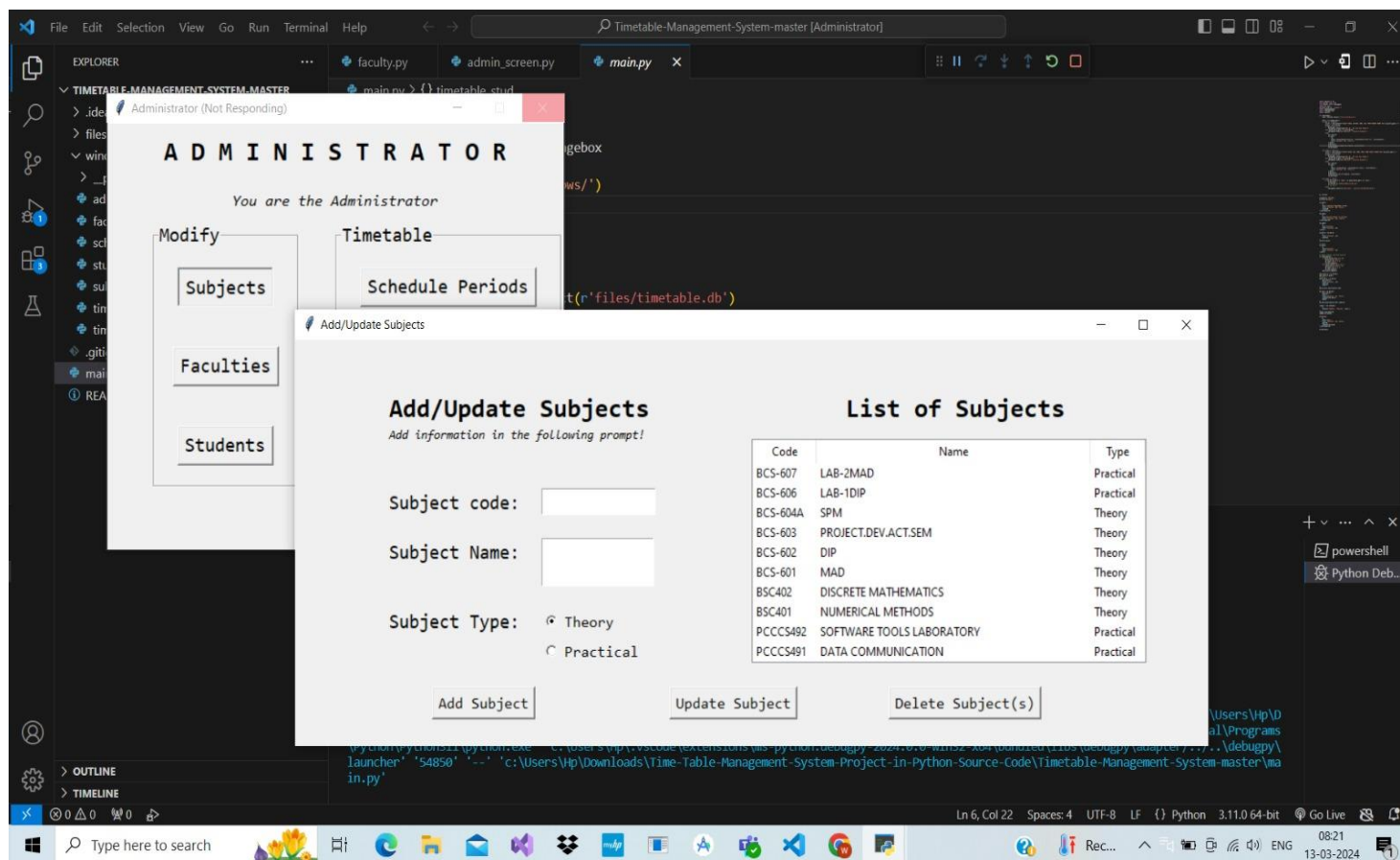
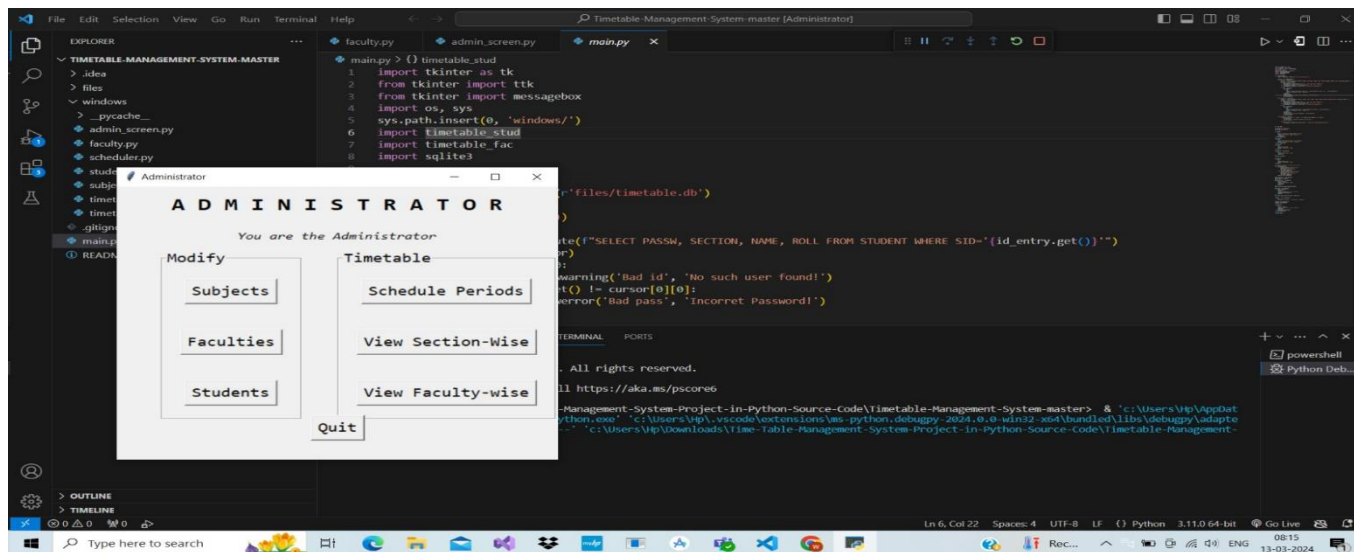
    m,
    values=['Student', 'Faculty', 'Admin']
)
combo1.pack(pady=15)
combo1.current(0)
```

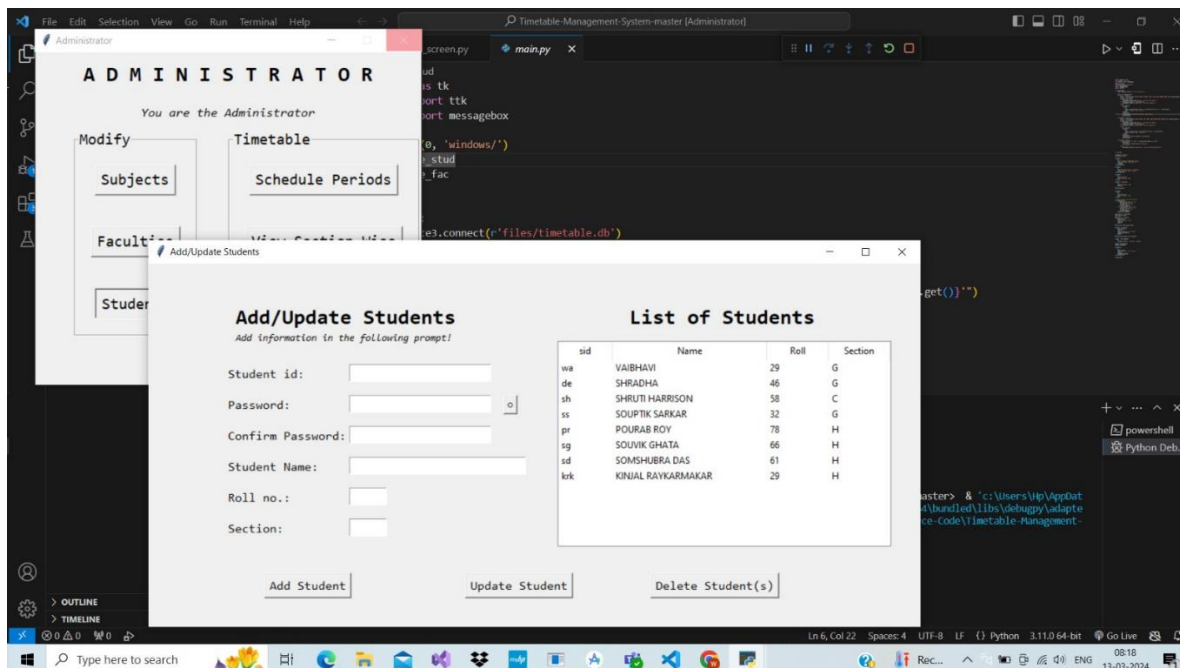
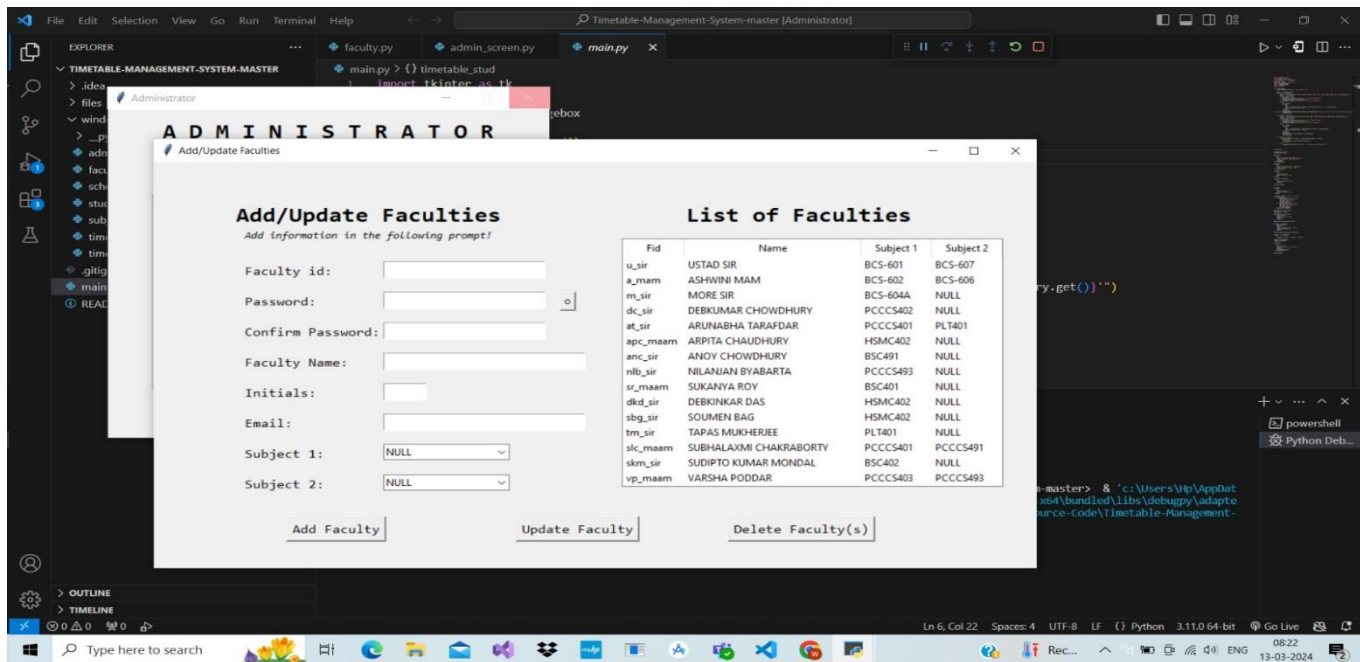


```
tk.Button(  
    m,  
    text='Login',  
    font=('Consolas', 12, 'bold'),  
  
    padx=30,  
    command=challenge  
).pack(pady=10)  
m.mainloop()
```

OUTPUT SCREENS :







tk

SHRADHA Section: G Roll No.: 46

T I M E T A B L E

Legend: Theory Classes Practical Classes

	Period 1	Period 2	Period 3		Period 4	Period 5	Period 6
Monday	BCS-602 A	BCS-601 U	BCS-604A H	R	No Class	No Class	No Class
Tuesday	BCS-602 A	BCS-601 U	BCS-604A H	E	No Class	No Class	No Class
Wednesday	BCS-602 A	BCS-601 U	BCS-604A H	C	BCS-606 A	BCS-607 U	No Class
Thursday	BCS-602 A	BCS-601 U	BCS-604A H	E	BCS-607 U	BCS-606 A	BCS-607 U
Friday	BCS-602 A	BCS-601 U	BCS-604A H	S	No Class	No Class	No Class

Type here to search

25°C 08:30 13-03-2024

tk

VAIBHAVI Section: G Roll No.: 29

T I M E T A B L E

Legend: Theory Classes Practical Classes

	Period 1	Period 2	Period 3		Period 4	Period 5	Period 6
Monday	BCS-602 A	BCS-601 U	BCS-604A H	R	No Class	No Class	No Class
Tuesday	BCS-602 A	BCS-601 U	BCS-604A H	E	No Class	No Class	No Class
Wednesday	BCS-602 A	BCS-601 U	BCS-604A H	C	BCS-606 A	BCS-607 U	No Class
Thursday	BCS-602 A	BCS-601 U	BCS-604A H	E	BCS-607 U	BCS-606 A	BCS-607 U
Friday	BCS-602 A	BCS-601 U	BCS-604A H	S	No Class	No Class	No Class

Type here to search

25°C 08:29 13-03-2024

TEACHER'S LECTURE TIMETABLE:

tk

USTAD SIR (U) Email: ustad123@gmail.com

T I M E T A B L E

Legend: Theory Classes Practical Classes

	Period 1	Period 2	Period 3		Period 4	Period 5	Period 6
Monday	No Class	Sections: 0	No Class	R	No Class	No Class	No Class
Tuesday	No Class	Sections: 0	No Class	E	No Class	No Class	No Class
Wednesday	No Class	Sections: 0	No Class	C	No Class	No Class	No Class
Thursday	No Class	Sections: 0	No Class	E	No Class	Sections: 0	No Class
Friday	No Class	Sections: 0	No Class	S	Sections: 0	No Class	Sections: 0
				S	No Class	No Class	No Class

Windows taskbar: Type here to search, 08:34, 13-03-2024

Future Improvement

This application is an answer for the time desk technology hassle manually. Its fundamental scope is to save the time and efforts for the time table generation process.

The statistics of the university with inside the facts based can similarly be used to keep a file of faculty's adventure for specific subject attribute correctness of undertaking will deliver a greater corrective method towards technology of this timetable.

This assignment will generate maximum corrective output and not using an errors. The destiny enhancement that may be developed from the undertaking is to generate the master timetable for the departments and to the entire university.

This enhancement may be carried out by making similar modifications maintaining the method and techniques used on this project.

CONCLUSION

In this paper we have introduced a model or a prototype for timetabling issues of middle scaled university in Sri Lanka. Even though the timetabling problem treated as an optimization issue, there is actually no fixed objective function to solve it. Therefore, after a proper literature survey, GA was selected to construction of course timetables developed for the University of Moratuwa.

This timetabling project seeks to generate near optimal timetables using the principles of genetic algorithm (selection, mutation and crossover) and it is easily understandable, less paper work, efficient and automated system, which helpful for authorities of the IT faculty. Major limitation of this TMSFIT are, the proposed system can only generate timetables based on a few hard constraints, it gives only optimal solutions not the best solution and it only generates timetables for courses and the execution time of GA itself is high. In future, this concept can be adapted to fit the construction of examination timetables also.

We suggest this timetabling system can be used as a framework and it will be more appropriate for medium scale universities

SUMMARY

This Time Table Management System is a quick and easy Python project for beginners and intermediates alike, and it broadens knowledge of Python applications

Finally, this entire Python project with free source code is an absolute project and a valuable means for users to learn and explore more about it.

Effective time management is all about achieving the right balance between your homework, university/college life, and your free time. As a student, organizing your days will eliminate stress and ensure that you are productive.

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